ROSSMANN

Sales forecasting

BY

- Aml Abdalluh
- Nada Yasser

Supervised By

- Mohammed Ammar

Agenda

Introduction 1

Description of the Dataset 2

Data Cleaning 3

Preprocessing 4

Exploratory Data Analysis (EDA) 5

Modeling 6

Results 7

Conclusion 8

Description of the Dataset

•Rossmann is one of the largest drugstore chains in Europe, headquartered in Germany.

•The Rossmann Store Sales dataset was originally released as part of a Kaggle competition.

•It contains real-world historical sales data from Rossmann stores. The dataset includes information such as:

•Sales figures (daily sales for each store).
•Store details (store type, assortment type, whether it runs promotions, etc.).
•Promotions and marketing campaigns (Promo, Promo2, and their intervals).
•Competition data (distance to the nearest competitor, when competitors opened nearby).
•Dates and holidays (to capture seasonality, weekends, and special events).

Description of the Dataset

- Dataset comes from retail stores
  Contains 4 main files: train.csv, test.csv, store.csv, sample_submission.csv
- Number of rows / records: ~1 million rows
  18 Features
- Features
  Numerical &Categorical &Dates
- Target Variable: Sales

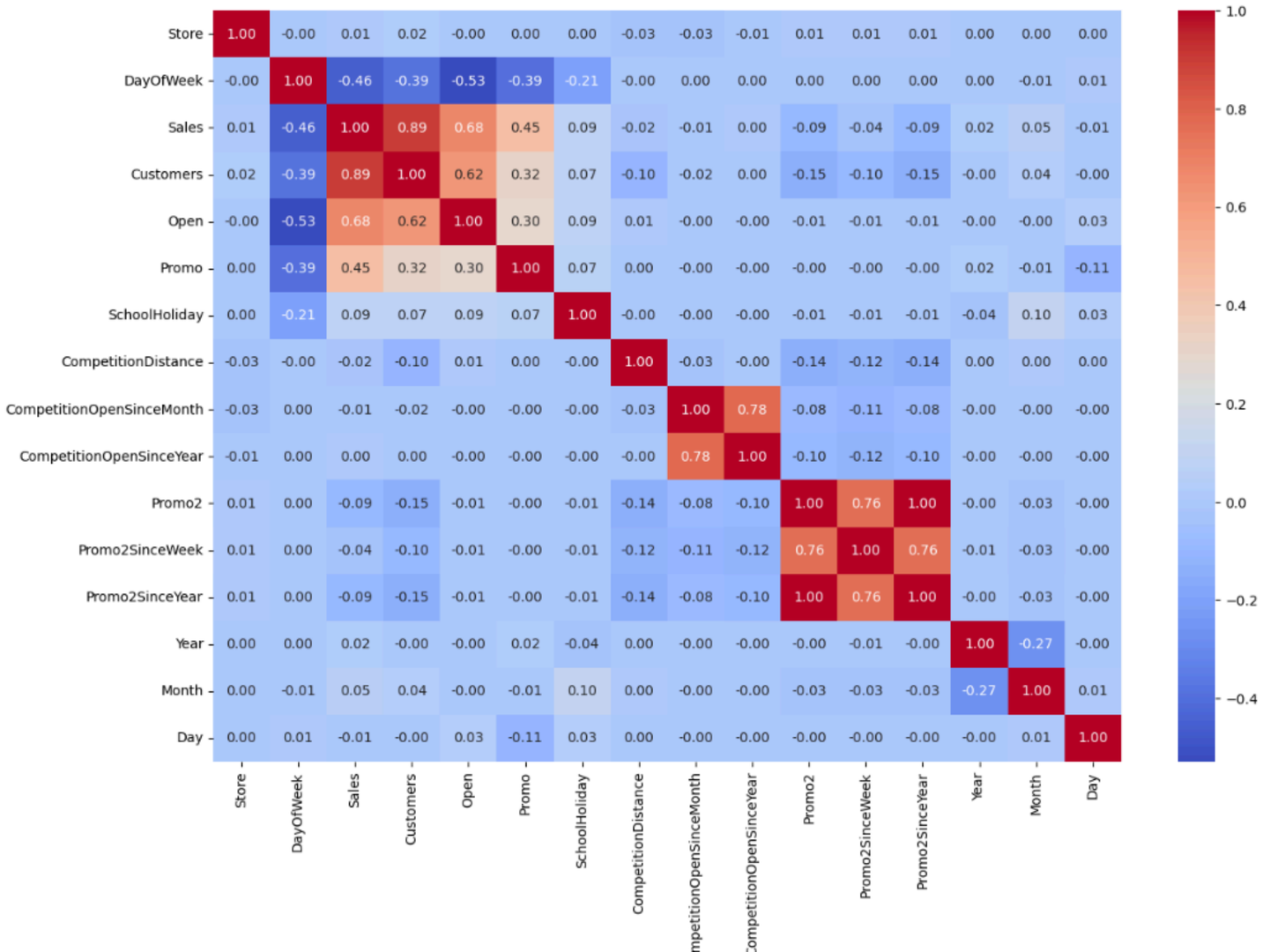
Numerical Values Check

	count	mean	std	min	25%	50%	75%	max
Store	1017209.0	558.429727	321.908651	1.0	280.0	558.0	838.0	1115.0
DayOfWeek	1017209.0	3.998341	1.997391	1.0	2.0	4.0	6.0	7.0
Sales	1017209.0	5773.818972	3849.926175	0.0	3727.0	5744.0	7856.0	41551.0
Customers	1017209.0	633.145946	464.411734	0.0	405.0	609.0	837.0	7388.0
Open	1017209.0	0.830107	0.375539	0.0	1.0	1.0	1.0	1.0
Promo	1017209.0	0.381515	0.485759	0.0	0.0	0.0	1.0	1.0
SchoolHoliday	1017209.0	0.178647	0.383056	0.0	0.0	0.0	0.0	1.0
CompetitionDistance	1014567.0	5430.085652	7715.323700	20.0	710.0	2330.0	6890.0	75860.0
CompetitionOpenSinceMonth	693861.0	7.222866	3.211832	1.0	4.0	8.0	10.0	12.0
CompetitionOpenSinceYear	693861.0	2008.690228	5.992644	1900.0	2006.0	2010.0	2013.0	2015.0
Promo2	1017209.0	0.500564	0.500000	0.0	0.0	1.0	1.0	1.0
Promo2SinceWeek	509178.0	23.269093	14.095973	1.0	13.0	22.0	37.0	50.0
Promo2SinceYear	509178.0	2011.752774	1.662870	2009.0	2011.0	2012.0	2013.0	2015.0

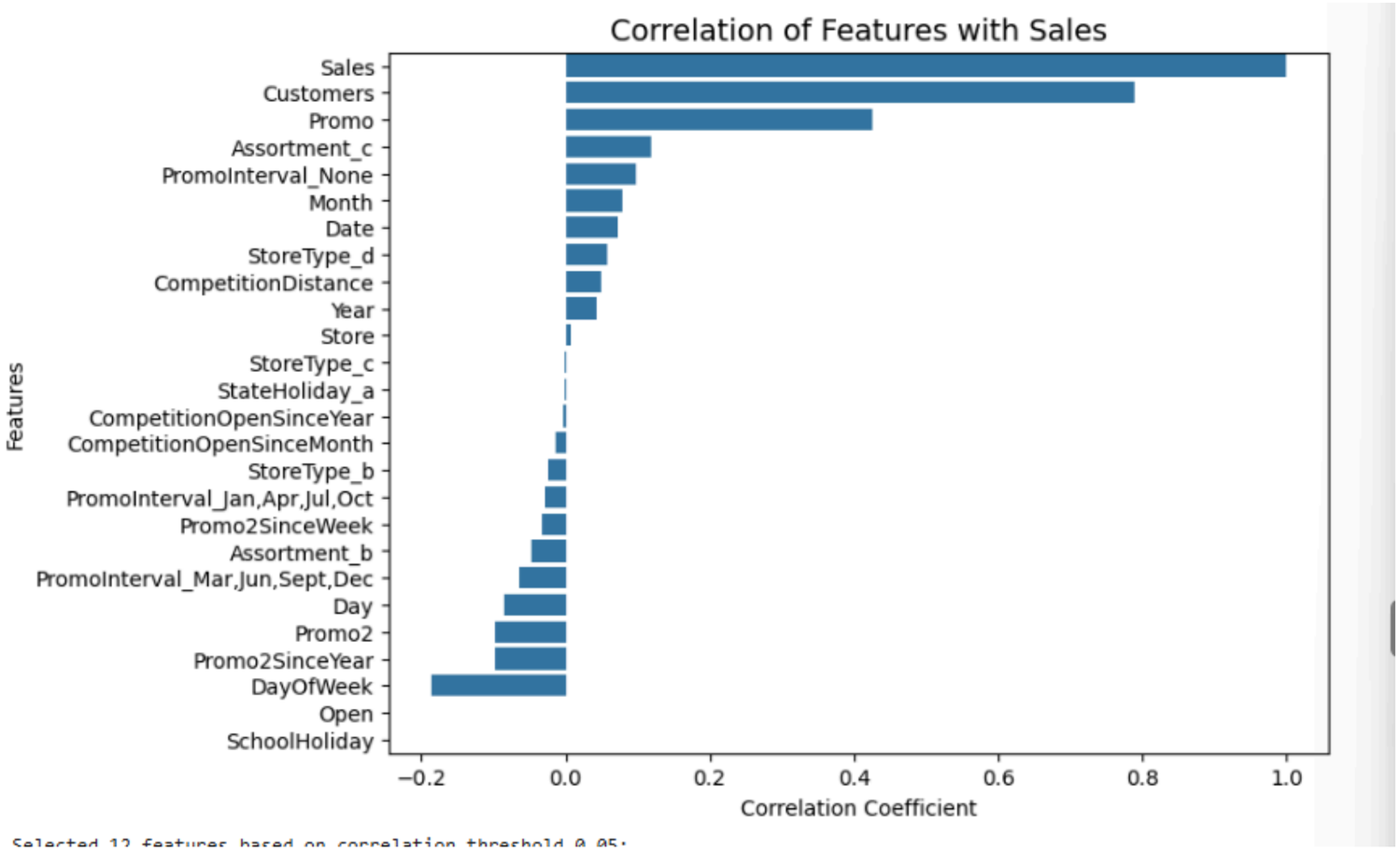
Categorical Values Check

	count	unique	top	freq
Date	1017209	942	2015-07-31	1115
StateHoliday	1017209	5	0	592943
StoreType	1017209	4	a	551627
Assortment	1017209	3	a	537445
PromoInterval	509178	3	Jan,Apr,Jul,Oct	293122

Relation Between Columns (Using Correlation Matrix)



Feature Importance



Data Cleaning & Preprocessing

HANDLING MISSING VALUES

CONVERTED DATE COLUMN TO DATETIME FORMAT

CATEGORICAL ENCODING

Missing values

Column	No. of missing values
CompetitionDistance	2642
CompetitionOpenSinceMonth	323348
CompetitionOpenSinceYear	323348
Promo2SinceWeek	508031
Promo2SinceYear	508031
PromoInterval	508031

Converted Date column to datetime format

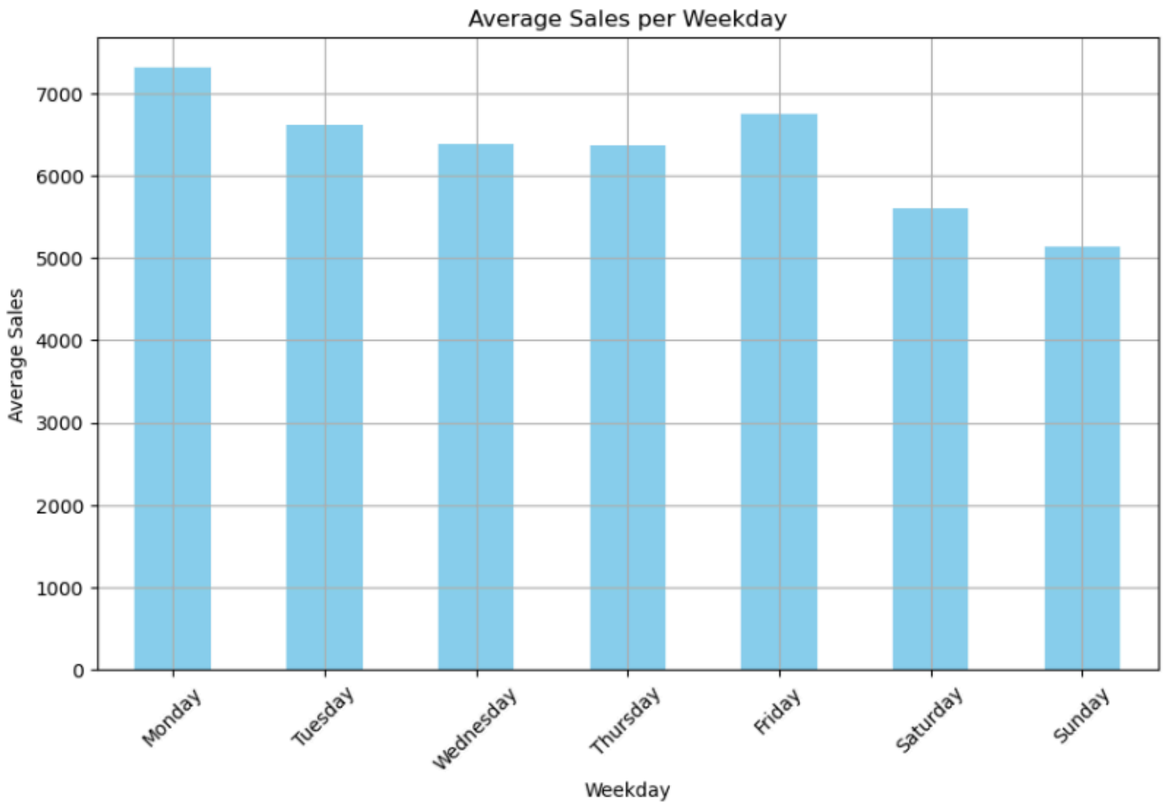
```
df["Date"] = pd.to_datetime(df["Date"])
df["Year"] = df["Date"].dt.year
df["Month"] = df["Date"].dt.month
df["Day"] = df["Date"].dt.day
df["DayOfWeek"] = df["Date"].dt.dayofweek
```

Categorical Encoding

```
df = pd.get_dummies(df, drop_first=True)
```

```
bool_cols = df.select_dtypes(include='bool').columns
df[bool_cols] = df[bool_cols].astype(int)
```

Exploratory Data Analysis (EDA)



Average Sales Analysis

Sales by Promotion:

Stores with promotions (Promo = 1) have higher average sales than stores without promotions.

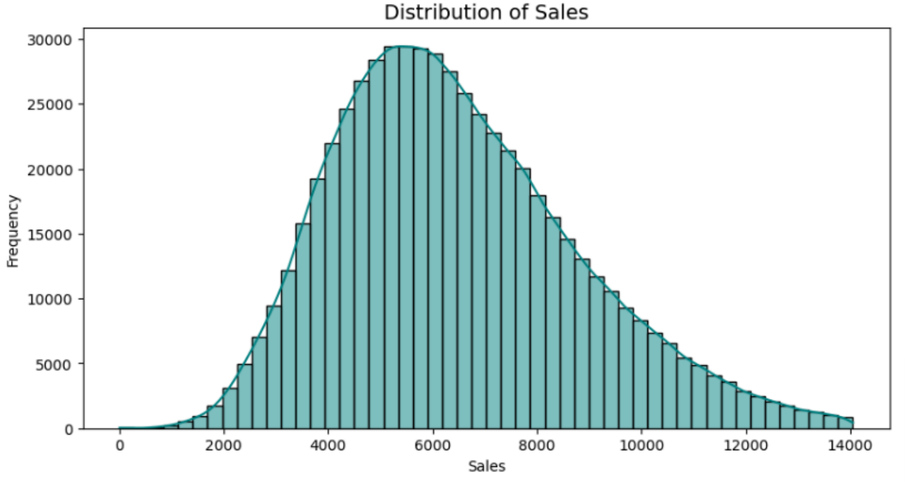
Sales by Day of Week:

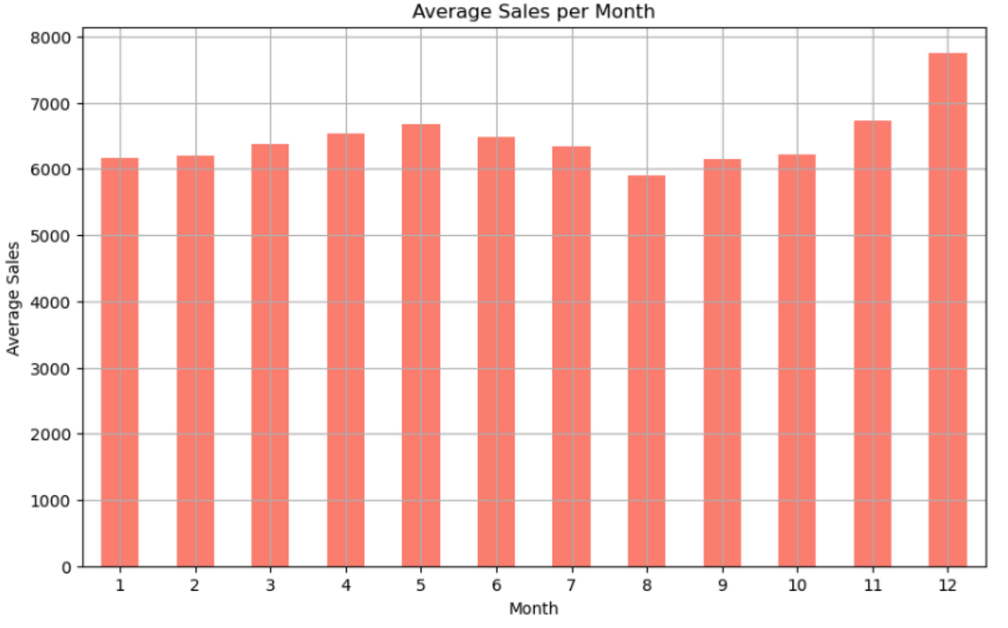
Sales tend to be higher on weekdays (Monday–Friday) and lower on weekends.

Sales by Store Type:

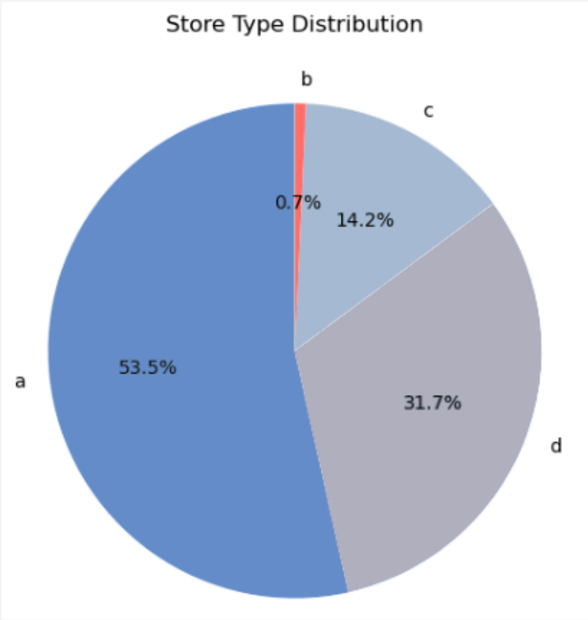
Different store types have different average sales, e.g., Type D > Type A > Type C > Type B.

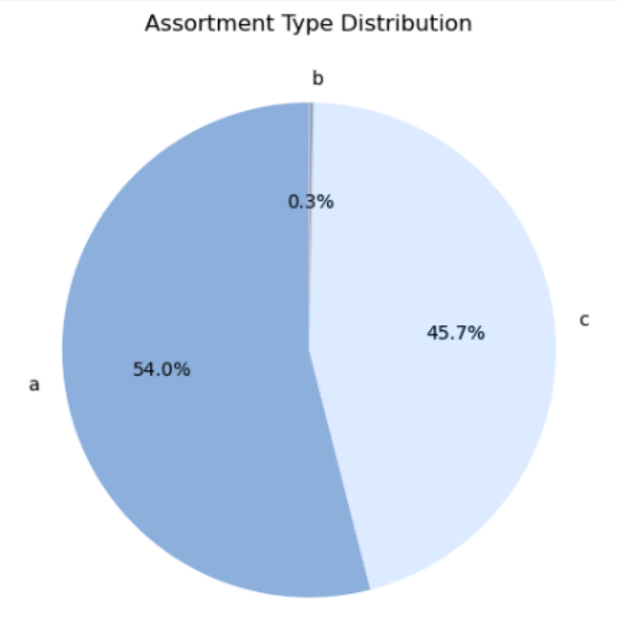
Store Activity by Day of Week
Number of Records
120000
100000
80000
60000
40000
20000
0
0
1
2
3
4
5
6
Day of Week (1=Mon, 7=Sun)

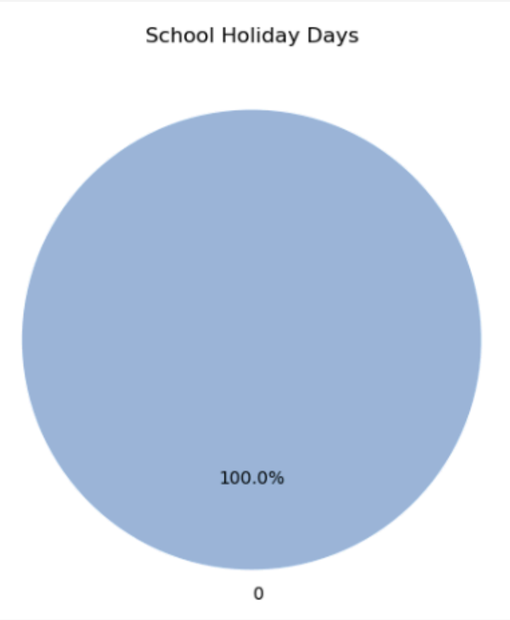

Distribution of Sales
Frequency
30000
25000
20000
15000
10000
5000
0
0
2000
4000
6000
8000
10000
12000
14000
Sales

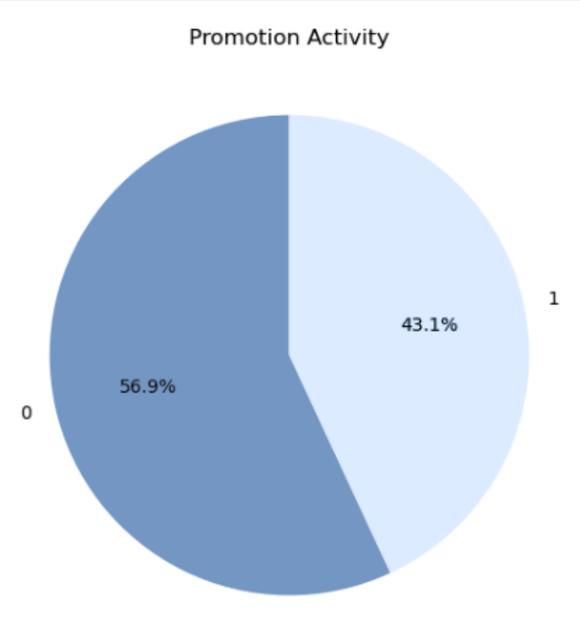

Average Sales per Month
Average Sales
8000
7000
6000
5000
4000
3000
2000
1000
0
1
2
3
4
5
6
7
8
9
10
11
12
Month

Pie Charts Insights









Modeling Alogritimes

Considered Models:

Model Type	Why Considered	Notes / Drawbacks
Linear Regression	Simple baseline, interpretable	Might underfit complex relationships
Random Forest	Handles non-linearities, robust	Can be slower with large datasets
KNN	Simple, no training time	Poor scalability with large datasets

Chosen Model:

Model Type	Why Considered	Notes / Drawbacks
✔ **XGBoost**	*High accuracy, handles missing data*	Efficient and interpretable

Data Splitting

- Split data into Training (80%) and Validation (20%)
- Validation set ensures true performance measurement and detects overfitting

Feature Engineering

- Extracted Month, Weekday, Year from Date column
- Created interaction features like Promo_Weekday and Promo_StoreType
- Helps the model capture seasonality and daily patterns in sales

Handling Missing Values

- Filled numerical columns with mean
- Filled categorical columns with mode
- Used training statistics only to avoid data leakage

Encoding Categorical Features

- Applied One-Hot Encoding for features like StoreType
- Applied Label Encoding for Weekday
- Converts categorical data into numeric form for XGBoost

Model Training (XGBoost)

- Chose XGBoost Regressor for tabular data
- Key hyperparameters:
  - n_estimators = 1200 → number of trees
  - max_depth = 8 → tree depth
  - learning_rate = 0.05 → step size
- Subsample and colsample help reduce overfitting

Evaluation

- Metrics used: MAE, RMSE, R^2
- Focused on Validation set for realistic performance
- Ensures the model generalizes to unseen data

Outlier Handling

- Applied log1p transformation to target (Sales) before training
- Reverted predictions using expm1 to get original sales values

Feature Importance

- Visualized top features with plot_importance
- Found Promo and DayOfWeek as major influencers of sales

Model Saving

- Saved trained model using joblib
- Included feature columns for consistent preprocessing during deployment
- Ensures safe usage in apps or APIs

Conclusion

- The model learned sales patterns effectively
- Achieved strong performance on unseen data
- Capable of generalizing and predicting future sales accurately

MODEL EVALUATION RESULTS

=== PERFORMANCE METRICS ===		
METRIC	TRAIN SET	VALIDATION SET
MAE (Mean Absolute Error)	0.04	0.04
RMSE (Root Mean Squared Error)	0.05	0.07
R^2 Score (Coefficient of Determination)	0.985	0.969

=== SAMPLE PREDICTIONS (TEST) ===	
SAMPLE #	PREDICTED VALUE
1	1.90
2	2.38
3	3.51
4	3.09
5	1.98

Note: Sample Predictions show specific output values for a subset of the test data.

Total Predictions Displayed: 5

```
Train RMSE: 609.3865803716348
Train R²: 0.9627656588221685
Validation RMSE: 721.520753789133
Validation R²: 0.9484021756482167
```

```
Train MAE: 645.8859021158619
Train RMSE: 941.1933522611074
Train R2: 0.9402674735026403
Validation MAE: 670.2523202523978
Validation RMSE: 988.7191947755955
Validation R2: 0.9338982270934085
Sample predictions on test: [1117.9324 1117.9324 1117.9324 1117.9324 1117
 1117.9324 1117.9324 1117.9324]
```

```
Test MAE: 298.0237117327774
Test RMSE: 444.999025939014
Test R2 Score: 0.9866098544611256
```

```
Train MAE: 290.49757650906156
Train RMSE: 424.15438902509635
Train R2: 0.9878688740170072
```

```
•••  Train MAE: 17.3805308267526
     Train RMSE: 72.48159488221457
     Train R2: 0.9996457509744674
     Validation MAE: 19.712697463973935
     Validation RMSE: 113.13181586498919
     Validation R2: 0.9991345605408223
```

```
Train MAE: 725.7622968353687
Train RMSE: 1067.3964473979333
Train R2: 0.9231746250824937
```

```
•••  Train RMSE: 554.81, R2: 0.943
     Validation RMSE: 607.48, R2: 0.932
     [4330.0317 7503.691  9224.212  6955.8765 7255.414 ]
```

```
Test MAE: 735.1396282340324
Test RMSE: 1087.4343276432448
Test R2 Score: 0.92003991378401
```

Thank You

NTI

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